Adapting Optimized AI Solution to Financial Fraud Detection

DENNIS J POTHIER

26 AUGUST 2025

Adaptation Strategies

The model developed for determining a health risk score based on other meteorological data could be adapted to the financial fraud detection domain with minimal structural modifications. The existing models' strengths in predictive forecasting could be very effective at predicting whether a fraudulent transaction is suspected to occur.

Strategies for adaptation include:

- Obfuscation of Any Personal Financial Data: Redaction or obfuscation of personally identifiable data must be performed before model training, which will likely be the first step.
- Target Variable Change: Replace the continuous regression label with a binary fraud indicator. (1 = Fraud, 0 = Legitimate)
- Domain-Specific Features: Inputs will shift from weather signals to transaction attributes such as distance from home, distance since last transaction, online purchase, etc. Feature engineering may be necessary to ensure the data tells a complete story. Data cleaning may also require different methods.
- Imbalanced Data Handling: The training loop must add weight to the fraudulent records for use in the model.
- Evaluation Metrics: The evaluation methods used, RMSE and R-squared, need to be replaced with classification metrics instead of the regression metrics. Measurements such as ROC-AUC, Precision-Recall AUC, precision, recall, and F1 are excellent measurements as they align with the high-risk, low-prevalence nature of fraud detection.

Rationale for Modifications

The domain-specific requirements drive the modifications, but some are needed to ensure model performance is as accurate as the weather prediction regression.

- Obfuscation of Any Personal Financial Data: Personal financial data is not necessary in training the model to perform, and by removing this from the dataset entirely, the model does not need as many legal requirements incorporated into it to ensure compliance.
- Target Variable Change: Fraud detection is inherently categorical. Either a record is flagged as fraud or not. Conversely, the weather dataset has continuous data for the labeled data. Reframing the objective is vital to ensuring model performance and aligning with operational needs.
- Domain-Specific Features: Transactional features provide a high predictive value, for example, a sudden large purchase in a foreign country with a recent transaction in a closer proximity to home is likely to be flagged as fraudulent. Feature engineering to capture specific predictors and cleaning will need to be performed to ensure data is normalized and does not overly affect model performance with the expected outliers in financial data. Creating new features to capture financial data in more discrete fashions may improve model performance.
- Imbalance Handling: Due to the typically sparse amount of fraudulent transactions in the typical real-world dataset, the model needs to focus more heavily on them. Without weighting adjustments, the model would overwhelmingly predict the legitimate class due to imbalance. Adding these mechanisms ensures that the rare but costly data is not overlooked.
- Metric Selection: Business stakeholders prioritize recall while tolerating some false positives in this case. False positives are preferable to false negatives, as one is just a

hassle for the consumer and the other is a considerable ordeal for them, not to mention the financial impact to the financial servicer by potentially refunding the transaction to the consumer and never regaining the funds.

Potential Impact

Adapting the model to financial fraud detection has significant operational and compliance implications to consider. Simply deploying the new, financially refined model is not without its hurdles. Things to consider:

- Scalability: Gradient boosting can efficiently scale to millions of transactions per day, but the optimized model also uses ensemble learning with other models. Dropping the final ensemble to simply using XGBoost may be necessary if using record sets of that scale, as many of the different models used in the ensemble do not perform as well with such a massive scale.
- Integration: The fraud model can be deployed using REST API or as an integration into real-time transaction monitoring systems—existing infrastructure transfers directly with minimal modifications.
- Performance: The model is expected to successfully recall fraudulent cases while maintaining acceptable precision, given that the modifications above are made to adjust for the class imbalance.
- Regulatory Compliance: This is the biggest concern of the model for actual adaptation to the financial sector. While whether related data is readily available and publicly usable, economic data is not. Ensuring to exclude, strip, or obfuscate personally identifiable information is vital to ensure that the system is auditable by the PCI DSS and GDPR regulatory frameworks.